

FPC & FFC Connector Selection Guide

A practical engineering reference for choosing the right flexible circuit connector — pitch, actuator type, contact position, and mechanical reliability.

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Interconnect Knowledge Base for Hardware Engineers

Contents

1. FFC vs. FPC: Knowing Which Technology You Need
2. Choosing an FPC Connector: A Four-Step Framework
3. ZIF vs. Non-ZIF: The Insertion Force Decision
4. Connector Pitch: Matching Density to Manufacturability
5. Designing the Flex Circuit for Long-Term Reliability
6. Quick-Reference Selection Checklist

This guide is a condensed reference drawn from the full FPC & FFC Connectors knowledge base at connectorselection.com/categories/fpc-ffc-connectors/, where each topic below is covered in greater depth with additional diagrams and edge cases.

1. FFC vs. FPC: Knowing Which Technology You Need

Flexible Flat Cables (FFC) and Flexible Printed Circuits (FPC) both route signals through space-constrained enclosures, but they are built differently and suit different jobs. Confusing the two often means over-engineering a simple jumper link, or under-specifying a complex signal path.

Flexible Flat Cable (FFC)

- **Construction:** straight, parallel flat copper conductors laminated between top and bottom insulation film (PET or polyimide).
- **Geometry:** strictly linear — all conductors run parallel at a fixed pitch (0.5mm, 1.0mm, or 1.25mm) end to end.
- **Tooling cost:** very low. Standard laminated stock is cut to length, stripped, and stiffened — no custom masks required.

Flexible Printed Circuit (FPC)

- **Construction:** copper foil on a polyimide substrate, chemically etched using the same photolithography as rigid PCBs.
- **Geometry:** supports non-linear shapes, branching arms, varying trace widths, SMT component pads, and controlled-impedance differential pairs.
- **Tooling cost:** high initial NRE — requires photomasks, laser-cut profiles, and custom lamination tooling.

How to Choose

- **Routing complexity:** straight point-to-point jumper → FFC. Branching, complex contours, or mounted components → FPC is required.
- **Unit cost at volume:** FFC is cheaper per unit at low-to-medium volumes — switching a simple interconnect from FPC to FFC can cut unit cost by up to 70%.
- **High-speed signal integrity:** FPC supports controlled impedance (50Ω single-ended, 100Ω differential) with ground planes, suited to PCIe, USB4, and MIPI CSI/DSI. Standard FFC has no ground plane or impedance control and is best kept to low-frequency signaling or power delivery.

2. Choosing an FPC Connector: A Four-Step Framework

Once FPC is the right cable technology, the connector itself has to be matched on four dimensions. Getting any one of these wrong shows up as cable pull-out, intermittent contact, or damaged contact beams during rework.

Step 1 — Actuator Type: ZIF vs. LIF

- **ZIF (Zero Insertion Force):** a movable latch opens the contact path so the cable slides in without friction; closing the latch compresses spring beams onto the pads. Flip-lock/back-flip actuators suit automated assembly; slider/push-pull actuators lock more robustly but need two-handed operation.
- **LIF (Low Insertion Force):** no moving actuator — the cable is pushed into a friction-fit slot. Lower cost and profile, but lower retention force and faster plating wear; best for low-pin-count, static connections.

Step 2 — Contact Position

- **Top contact:** spring beams on the housing roof — cable pads must face up.
- **Bottom contact:** spring beams on the housing floor — cable pads must face down.
- **Dual contact:** spring beams on both walls — accepts either orientation, removing assembly-line errors and adding redundancy under vibration.

Step 3 — Stiffener Thickness Match

FPC connectors are engineered for a specific total cable-end thickness — most commonly $0.30\text{mm} \pm 0.03\text{mm}$ or $0.20\text{mm} \pm 0.03\text{mm}$. Too thin, and contacts can't generate enough normal force (intermittent connection); too thick, and forcing the cable in bends or breaks the internal contact beams.

Step 4 — Mechanical Retention

- **Side catcher / tab locks:** plastic ears or metal latches that engage molded notches on the FPC, resisting axial pull-out once the actuator closes.
- **Anchoring SMT solder tabs:** wide, robust solder tabs on both ends of the housing that anchor it to the PCB and absorb peel forces from cable flexure.

3. ZIF vs. Non-ZIF: The Insertion Force Decision

This is the single mechanical decision with the biggest impact on both assembly process and long-term reliability, so it earns its own closer look.

	Non-ZIF	ZIF
Mechanism	Fixed cavity with spring-loaded contacts; cable is pushed straight in against contact resistance.	Movable actuator (flip-lock or slider) opens the contact path for friction-free insertion, then locks to apply uniform force.
Wear mechanism	Insertion friction scrapes plating off the cable traces — limits the interface to a small number of mating cycles.	No insertion friction; wear is minimized across many mate/unmate cycles.
Best fit	Low-cost, low-pin-count (roughly under 20 positions), one-time factory assembly.	High-density, fine-pitch (0.3mm/0.2mm) interfaces, or any connection expected to be mated/unmated repeatedly.
Moving parts	None — cheaper, and immune to actuator breakage from drops.	Actuator adds a failure point but is required once pin count and pitch make simultaneous friction-fit insertion unviable.

Rule of thumb: Non-ZIF wins on cost and simplicity for permanent, factory-only connections. ZIF wins on reliability and trace preservation anywhere the interface is high-density, fine-pitch, or expected to be mated more than once.

4. Connector Pitch: Matching Density to Manufacturability

Pitch — the center-to-center spacing between pins — sets both the connector's footprint and how difficult it is to manufacture reliably. As designs shrink, pitch has moved from 1.0mm/0.5mm standards down to 0.4mm, 0.3mm, and even 0.2mm.

Pitch	Typical Use	Current / Contact	Manufacturing Note
0.5mm	Display modules, internal jumpers, automotive infotainment	0.5–1.0A	Standard SMT tolerances; no HDI required
0.4mm	Smartphones, smartwatches, compact camera modules	0.3–0.5A	Needs tight stencil control to avoid solder bridging
0.3mm / 0.2mm	Wearables, AR/VR, hearing aids, ultra-compact sensors	0.2–0.3A	Requires HDI fabrication and optical assembly alignment

Design Factors That Matter More Than Pin Density Alone

- **Assembly yield:** a finer pitch saves board area but widens the risk of solder bridging and rework. Default to 0.5mm or 0.4mm unless space strictly requires 0.3mm.
- **Power rail allocation:** power delivered over a fine-pitch connector needs multiple parallel contacts (often 5–8 for a display backlight rail), which erodes much of the space savings.
- **Stiffener alignment:** below 0.4mm pitch, small shifts in stiffener lamination during fabrication can misalign cable pads against the connector's spring contacts.

5. Designing the Flex Circuit for Long-Term Reliability

Choosing the right connector only solves half the problem — the flex circuit itself has to be laid out to survive assembly and field flexing. Standard rigid-PCB layout habits, applied directly to a flex circuit, tend to produce cracked traces, coverlay delamination, and stiffener detachment.

Bend Radius

- **Static (flex-to-install):** bent once during assembly — keep minimum bend radius at 6–10× total cable thickness.
- **Dynamic (continuous flexing):** bent repeatedly in use (hinges, moving heads) — keep minimum bend radius at 20–30× thickness, and use rolled-annealed (RA) copper rather than electro-deposited (ED) copper.

Trace Layout in the Flex Region

- **Stagger multi-layer traces:** offsetting top/bottom traces avoids the "I-beam effect" that stiffens the substrate and concentrates bending stress.
- **Avoid sharp trace corners:** use smooth radiused arcs through any bend zone — 90°/45° corners concentrate stress and tear.
- **Add tear-stops and fillets:** radius internal corners at least 0.75–1.0mm, and place copper tear-stop rings at slot roots to arrest tearing.

Stiffener and Coverlay Integration

- **Rigid stiffener at the contact zone:** laminate polyimide or FR4 behind the contact fingers to hold the connector's required thickness (commonly 0.30mm ±0.03mm).
- **Overlap stiffener and coverlay boundaries:** never let the two edges land on the same line — overlap by 0.5–1.0mm to avoid a sharp stress hinge where traces snap.
- **Strain relief adhesive:** a UV-cure epoxy or RTV silicone bead at the rear of the connector slot absorbs flexure before it reaches the SMT solder joints, important in high-vibration designs.

6. Quick-Reference Selection Checklist

Use this as a final pass before locking in a connector and cable spec.

- Confirmed FFC vs. FPC based on routing complexity, volume, and signal speed
- Actuator type chosen (ZIF for fine-pitch or repeated mating, LIF/Non-ZIF for low-cost one-time assembly)
- Contact position (top / bottom / dual) matched to cable pad orientation
- Stiffener thickness spec confirmed against connector datasheet tolerance
- Mechanical retention features specified for the vibration/drop environment
- Pitch selected against assembly yield and power-rail contact count, not density alone
- Bend radius rule applied (static: 6–10× thickness / dynamic: 20–30× thickness)
- Stiffener and coverlay boundaries overlapped by 0.5–1.0mm
- Strain relief adhesive specified for high-vibration applications

Want a second opinion on a specific design? Submit your question through [Ask the Engineer at connectorselection.com/ask-the-engineer/](https://connectorselection.com/ask-the-engineer/) — every question is reviewed, and the best get answered publicly.

This guide condenses material from the FPC & FFC Connectors knowledge base at connectorselection.com. Values shown are general engineering guidance — always verify against your specific connector manufacturer's datasheet before finalizing a design.